

House Price Prediction

Advanced Regression on the Ames, Iowa Housing Dataset

Project Report — Data Cleaning, EDA, Feature Engineering, Modeling,
Tuning & Business Value

0.906

R^2 – XGBoost

0.1199

5-fold CV RMSE (log)

80 → 287

raw → engineered features

1. Business Problem

A real-estate company wants to estimate the selling price of a house **before** it is listed. A reliable valuation model lets the business set competitive prices, sell faster, and avoid the cost of over- or under-pricing inventory.

Business benefits

- **Better pricing strategy** — data-driven list prices instead of guesswork.
- **Faster sales** — correctly priced homes spend less time on market.
- **Reduced over/under-pricing risk** — fewer mark-downs and missed margin.
- **Smarter investment decisions** — spot undervalued properties to acquire.
- **Accurate valuation** of future developments and portfolios.

Target variable: SalePrice (modeled on the log scale to handle right-skew).

2. The Dataset

The Ames Housing dataset (De Cock, 2011) contains **1,460 houses** described by **80 features** plus the SalePrice target. Features span every dimension a buyer cares about:

Category	Example features
Area	LotArea, GrLivArea, TotalBsmtSF
Quality	OverallQual, OverallCond, KitchenQual
Location	Neighborhood, Condition1
Age	YearBuilt, YearRemodAdd, YrSold
Garage	GarageArea, GarageCars, GarageType
Basement	TotalBsmtSF, BsmtFinSF1, BsmtQual

3. Exploratory Data Analysis

EDA shaped every downstream decision. Six analyses stood out:

3.1 Target distribution

SalePrice is strongly right-skewed — a few luxury homes create a long tail. Applying **log1p(SalePrice)** produces a near-normal target, which stabilizes regression error and is the metric Kaggle scores on.

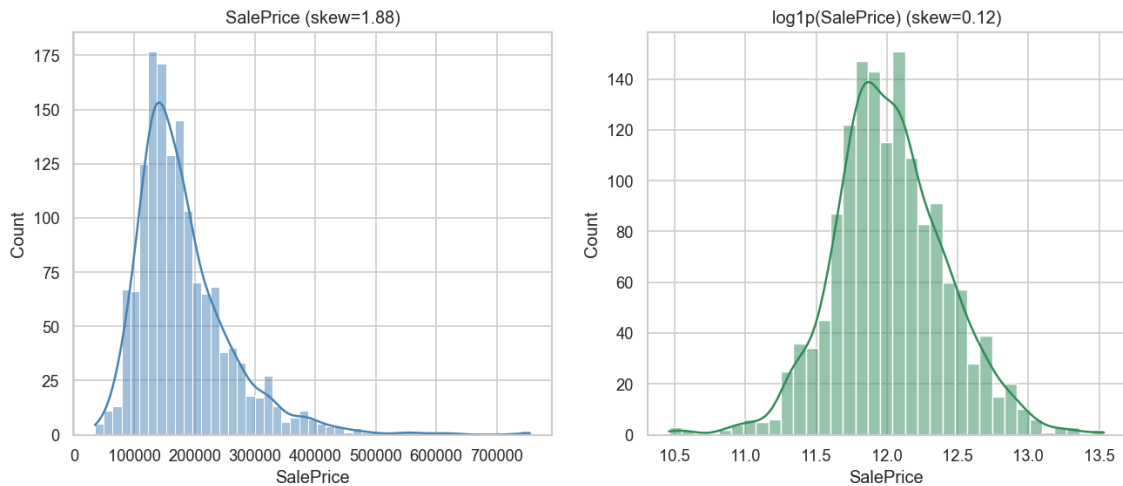


Fig 1. Raw SalePrice (left) vs log-transformed (right).

3.2 What correlates with price?

Quality and size dominate. OverallQual (~0.79), GrLivArea (~0.71), GarageCars/Area and TotalBsmtSF show the strongest correlations with SalePrice.

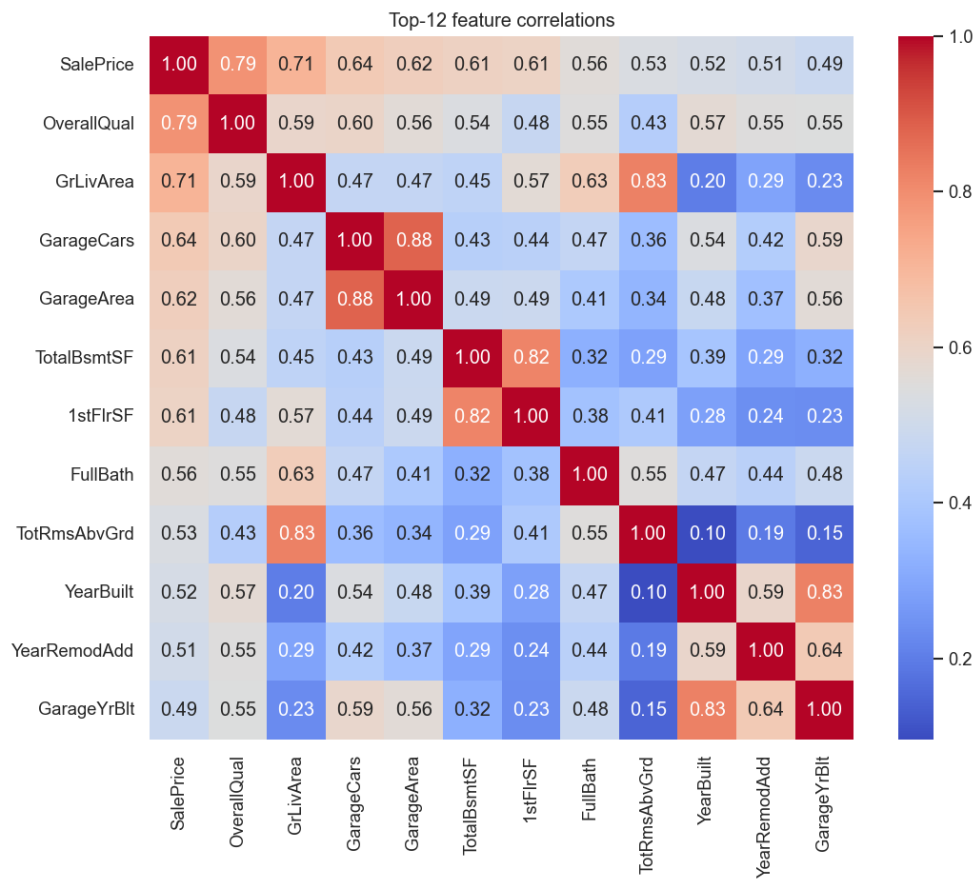


Fig 2. Correlation heatmap of the top features.

3.3 Living area & outliers

Price rises with living area, but two very large homes sold cheaply (documented partial sales). These outliers were removed so they don't distort the fit.

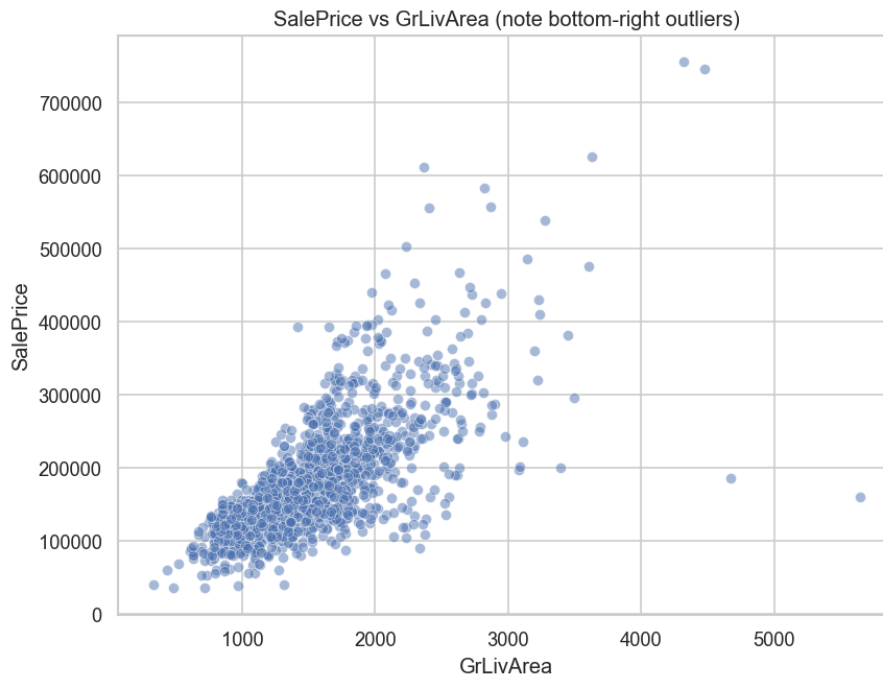


Fig 3. SalePrice vs GrLivArea — outliers bottom-right.

3.4 Location, location, location

Median price varies **3–4×** across neighborhoods — location is one of the most important drivers of value.

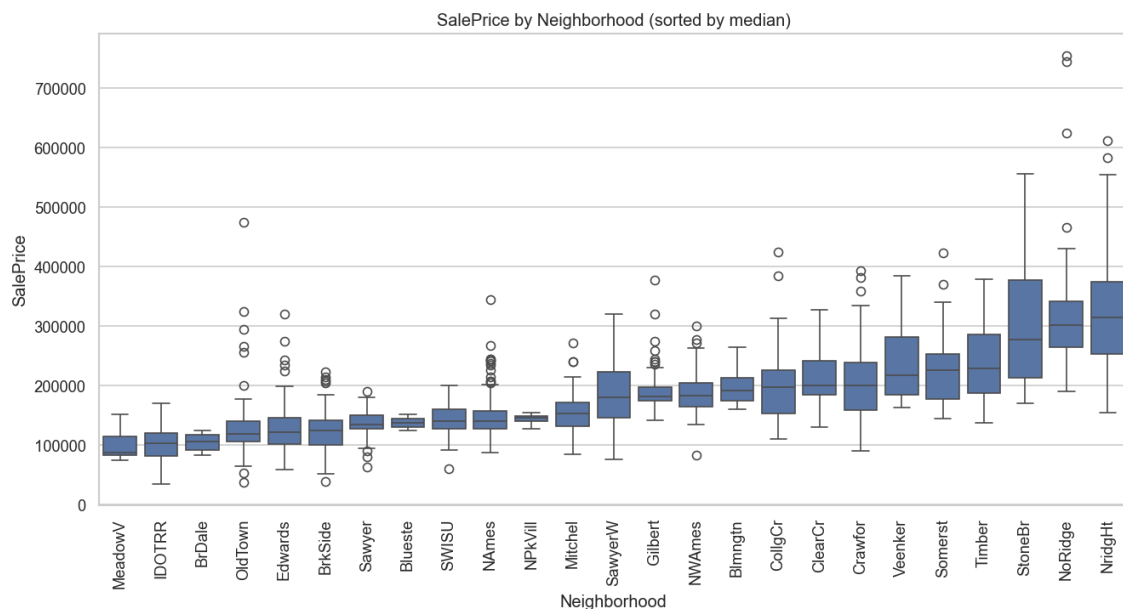


Fig 4. SalePrice by neighborhood (sorted by median).

3.5 House age

Newer houses command a premium; price declines with age, motivating engineered age features.

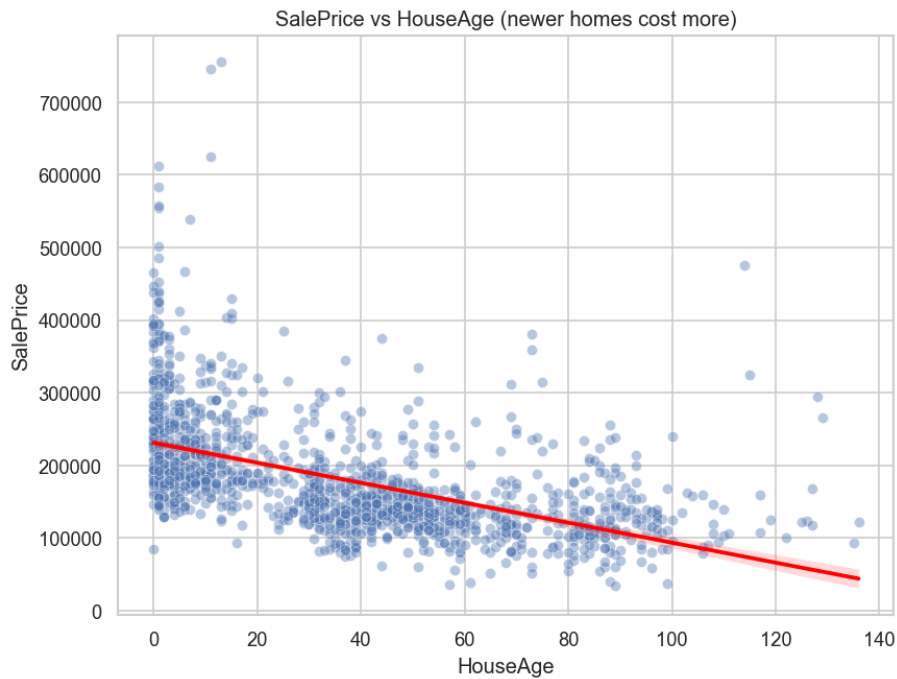


Fig 5. SalePrice vs house age.

3.6 Missing data

Missingness is concentrated in Pool, MiscFeature, Alley, Fence and Garage/Basement fields — where a blank usually means the feature is simply absent, not unknown.

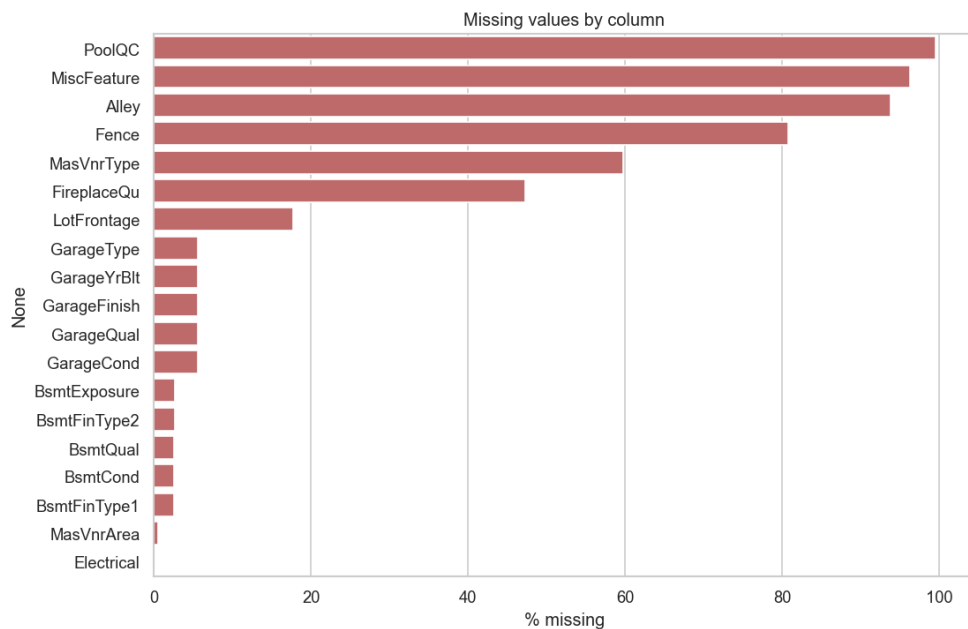


Fig 6. Percentage of missing values by column.

4. Data Cleaning

Missing values were imputed with **domain-aware** rules rather than blind defaults:

- **"None"** for categoricals where a blank means the feature is absent (PoolQC, GarageType, Fence, BsmtQual, MasVnrType...).
- **0** for the matching numeric fields (GarageArea, TotalBsmtSF, MasVnrArea, BsmtFullBath...).
- **Neighborhood median** for LotFrontage (lot frontage tracks the local block layout).
- **"Typ"** for Functional, per the data dictionary.
- **Mode / median** for any remaining categorical / numeric gaps.
- Removed the **2 documented GrLivArea outliers** (>4000 sqft, low price).

5. Feature Engineering

New features encode the intuition that **age, total size and quality** drive price. One-hot encoding then expanded the matrix from 80 raw to 287 model-ready features.

Feature	Definition	Why it helps
HouseAge	YrSold - YearBuilt	Newer = premium
RemodAge	YrSold - YearRemodAdd	Recent reno adds value
TotalSF	Bsmt + 1st + 2nd floor	Single size signal
TotalArea	GrLivArea + TotalBsmtSF	Usable space
TotalBath	Full + 0.5*Half (+bsmt)	Combined bathrooms
TotalPorchSF	Sum of all porch areas	Outdoor space
QualScore	OverallQual * OverallCond	Quality interaction
Has* flags	Pool/Garage/Bsmt/2nd/Fireplace	Presence signals

6. Modeling & Hyperparameter Tuning

Four models were trained on an 80/20 split and validated with **5-fold cross-validation** on the log target:

- **Linear Regression** (scaled) — interpretable baseline.
- **Random Forest** — captures non-linearity and interactions.
- **XGBoost** — gradient boosting, strong on tabular data.
- **LightGBM** — fast histogram-based boosting.

Tuning approach

Each tree model was tuned with **RandomizedSearchCV** (25 iterations, 3-fold inner CV) — faster than exhaustive grid search while covering a wide space:

- **Random Forest:** `n_estimators`, `max_depth`, `min_samples_split/leaf`, `max_features`.
- **XGBoost:** `learning_rate`, `max_depth`, `subsample`, `colsample_bytree`, `gamma`, `reg_lambda`.
- **LightGBM:** `num_leaves`, `learning_rate`, `max_depth`, `subsample`, `colsample`, `min_child_samples`.

Cross-validation gives a more reliable score than a single split and guards against overfitting during model selection.

7. Results & Model Comparison

Model	R ²	RMSE	MAE	CV RMSE	CV std
XGBoost	0.9058	0.1260	0.0833	0.1199	0.0086
LightGBM	0.8979	0.1312	0.0857	0.1246	0.0063
RandomForest	0.8736	0.1460	0.0974	0.1341	0.0100
LinearRegression	0.8838	0.1399	0.0933	0.1362	0.0164

Winner: XGBoost with $R^2 = 0.906$ and 5-fold CV RMSE = 0.1199 on the log target. Gradient boosting beats the linear and bagging baselines, as expected for structured tabular data.

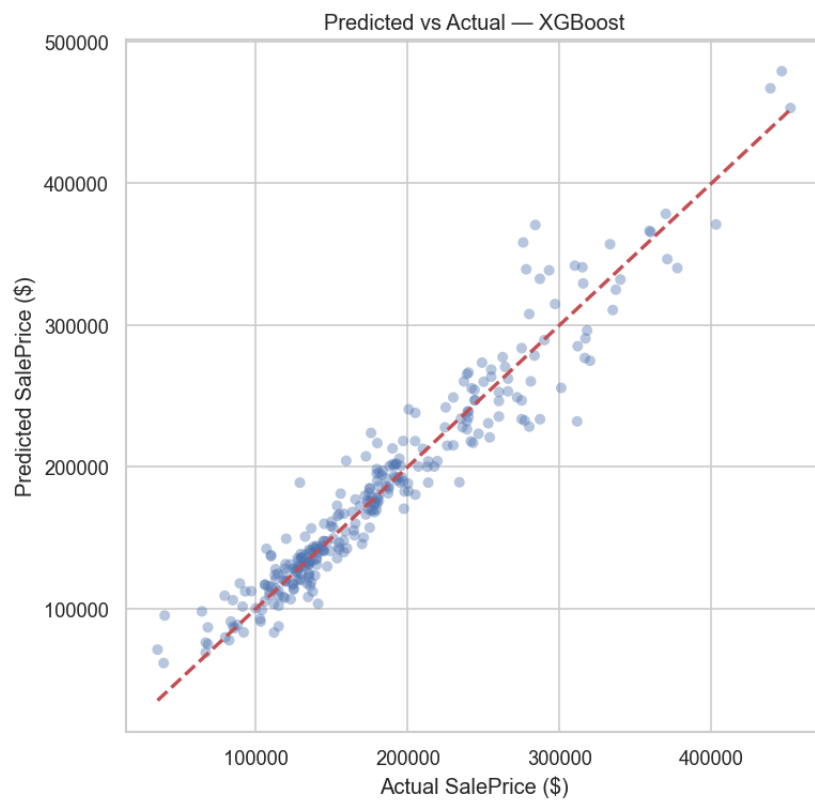


Fig 7. Predicted vs actual SalePrice for the best model (points hug the 45° line).

8. Explainability (SHAP)

SHAP values explain *why* the model predicts a given price, turning a black box into actionable insight. The most influential features are consistently:

- **OverallQual** — overall material & finish quality.
- **GrLivArea / TotalSF** — total living space.
- **Neighborhood** — location premium.
- **TotalBsmtSF, GarageCars** — basement & garage capacity.
- **YearBuilt / HouseAge** — newer homes priced higher.

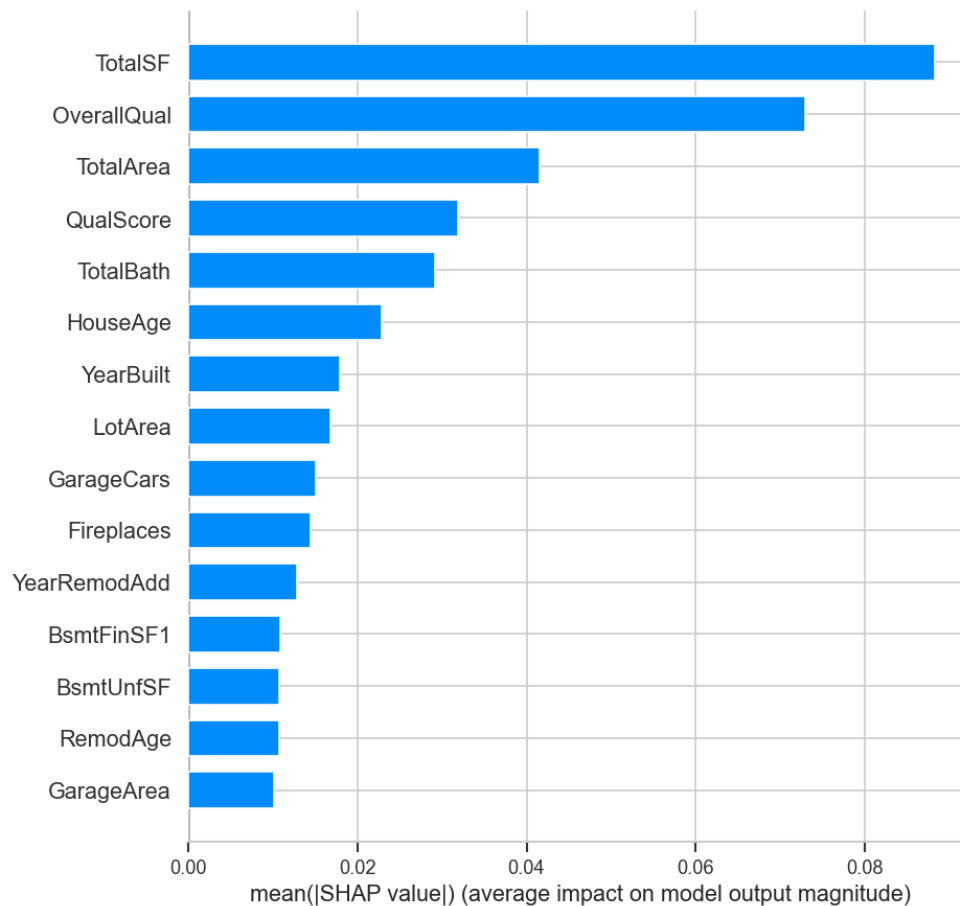


Fig 8. Mean |SHAP| feature importance.

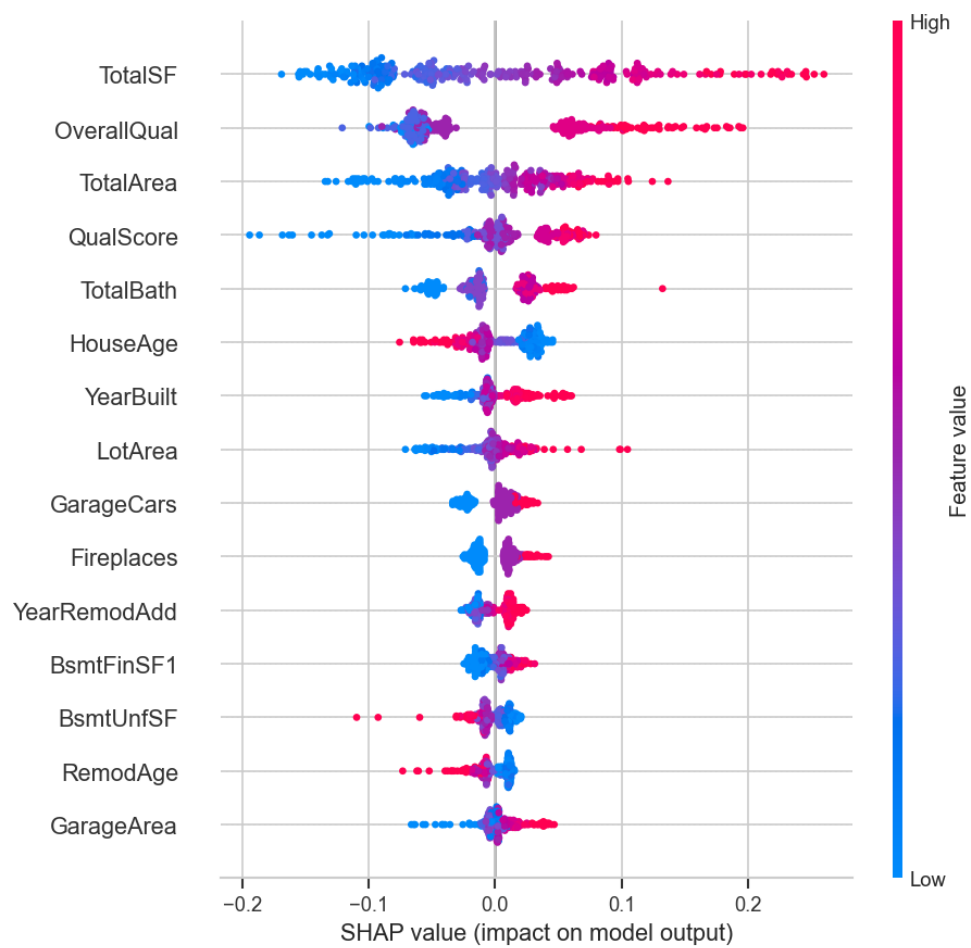


Fig 9. SHAP summary — feature value vs price impact.

9. Business Value

The model converts raw property attributes into a defensible price estimate that the business can act on:

- **Pricing engine:** an instant valuation for any new listing within ~12% RMSE on the log scale — tight enough to anchor list price.
- **Faster turnover:** accurate prices reduce time-on-market and costly mark-downs.
- **Acquisition screening:** flag homes priced below model value as investment candidates.
- **Renovation ROI:** SHAP shows quality and finished area move price most — guiding where upgrade spend pays back.
- **Location intelligence:** neighborhood ranking informs where to buy and build.
- **Transparency:** SHAP explanations make every valuation auditable for agents and clients.

Key takeaways

- Quality drives price more than any single size metric.
- Location can swing value 3–4× between neighborhoods.
- Bigger, newer, well-finished homes sell for more — quantifiably.

10. Future Improvements

- **Stacked ensemble** of XGBoost + LightGBM + Random Forest (+1–3%).
- **External data:** school ratings, crime rate, distance to city center, economic indicators.
- **Deep learning** tabular baseline for comparison.
- **BI dashboard** (Power BI / Tableau): price distribution, neighborhood ranking, predicted-vs-actual, top drivers.
- **Deployment:** wrap the saved model behind an API for real-time valuation.

Generated from project artifacts in `outputs/`. Reproduce with `python main.py` then `python make_report.py`.